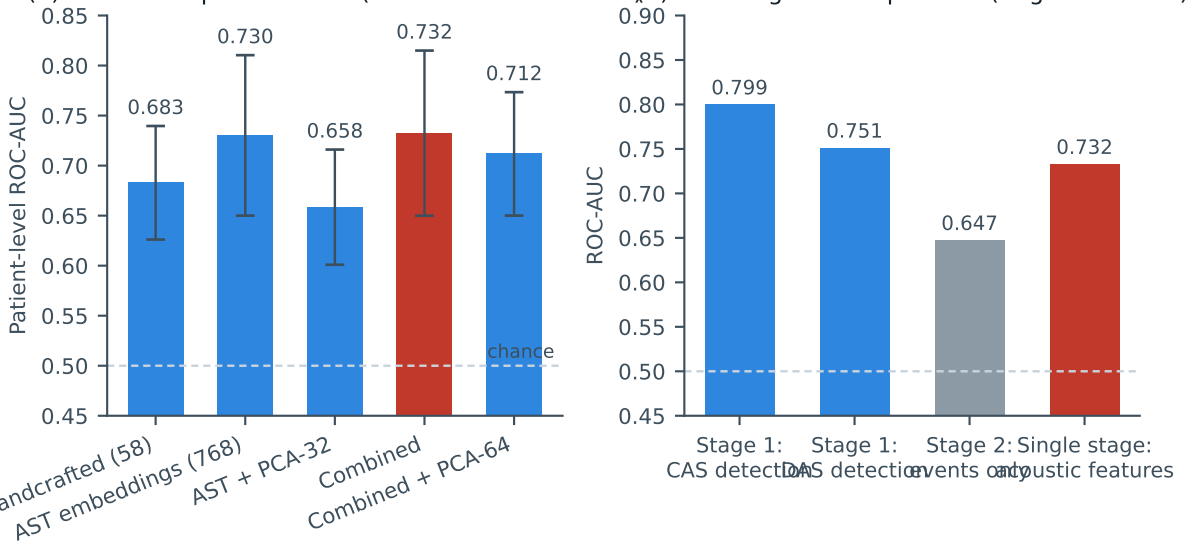
(a) Acoustic representation (mean ± SD over folds)
Patient-level ROC-AUC
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
0.683
0.730
0.658
0.732
0.712
chance
Handcrafted (58)
AST embeddings (768)
AST + PCA-32
Combined
Combined + PCA-64
(b) Two-stage decomposition (negative result)
ROC-AUC
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
0.799
0.751
0.647
0.732
Stage 1: CAS detection
Stage 1: DAS detection
Stage 2: events only
Single stage: acoustic features